

Project 02

AI Music & Audio Analyzer

Test & Development Report — v1.0

Item	Current Value
Project Status	Milestone v1.0 — Steps 1–18B validated
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
DSP Milestone	Steps 1–17 complete
AI Milestone	Step 18A + Step 18B complete
Audio Health Score	96 / 100 — Excellent

1. Project Overview

Project-02 is a practical Python-based AI Music & Audio Analyzer developed using a Windows laptop and Airdopes Joy Bluetooth earbuds. The project evolved from Bluetooth audio-device validation into a DSP pipeline covering waveform, frequency, spectral, rhythm, musical-key, loudness and clipping analysis. Version 1.0 adds an AI interpretation layer that converts measured audio features into structured, human-readable music analysis.

2. v1.0 Milestone

Version 1.0 marks the first complete AI-assisted analysis milestone. The system can perform measured DSP analysis, generate a transparent audio health score, package measurements as JSON, and use Gemini AI to interpret those measurements without requiring the AI to invent unsupported track metadata.

3. End-to-End Architecture

Airdopes / Music File → Audio Loading → Signal Analysis → Waveform → FFT → Spectrogram → Spectral Features → Bass/Mid/Treble → Tempo/Beat → Integrated Summary → Key/Chroma → Loudness/Dynamic Range → Clipping → Audio Health Score → Feature JSON → Gemini AI → Structured AI Interpretation

4. Completed Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS
15	Key / Chroma Analysis	PASS
16	Loudness / Dynamic Range / Clipping	PASS
17	Audio Quality & Health Score	PASS
18A	Music Feature Package	PASS
18B	AI Music Interpretation	PASS

5. Hardware Validation

Parameter	Result
Bluetooth Device	Airdopes Joy
Microphone Mode	Hands-Free AG Audio
Input Channels	1
Sample Rate	44,100 Hz
Test Recording	5.00 sec
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638
Signal Classification	Healthy

6. Real Music Track Baseline

Parameter	Result
File	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
Samples	10,585,088
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Bass / Mid / Treble	89.42% / 10.50% / 0.08%
Dominant Band	Bass
Estimated BPM	109.96
Detected Beats / Onsets	437 / 1,052

7. Key / Chroma Analysis

Feature	Result
Dominant Pitch Class	D
Estimated Key	G Minor
Best Correlation	0.7472
Second Candidate	D Minor
Second Correlation	0.7196

Feature	Result
Confidence Gap	0.0276
Reported Confidence	High
Output	reports\music_key_chroma.png

G Minor is the strongest candidate from chroma/profile correlation. D Minor remains a close alternative, so the result is documented as an estimated key rather than ground truth.

8. Loudness, Dynamic Range & Clipping

Parameter	Result
RMS Amplitude	0.140450
RMS Level	-17.05 dBFS
Peak Amplitude	0.980979
Peak Level	-0.17 dBFS
Crest Factor	6.985
Peak-to-RMS Dynamic Range	16.88 dB
Average Short-Term RMS	-18.07 dBFS
Maximum Short-Term RMS	-10.10 dBFS
Minimum Short-Term RMS	-102.59 dBFS
Short-Term Level Variation	92.49 dB
Clipping Threshold	0.999
Total Clipped Samples	0
Clipping Percentage	0.000000%
Clipping Status	No Clipping Detected
Output	reports\music_loudness_dynamic_range.png

The 92.49 dB short-term variation is strongly influenced by very low-level/silent frames and is not treated as a standardized music dynamic-range measurement. The 16.88 dB peak-to-RMS value is retained as a simple signal-level dynamic indicator.

9. Audio Quality & Health Score

Component	Score	Maximum
Signal Level	20	20
Clipping Safety	20	20
Dynamic Characteristics	15	15
Frequency Balance	11	15
Spectral Quality	10	10
Rhythm	10	10
Key / Chroma	10	10
TOTAL	96	100

Final project health classification: **96 / 100 — Excellent**. This is a transparent, project-specific engineering heuristic and not an industry-standard mastering, LUFS, codec-quality or perceptual-quality score.

10. Step 18A — Music Feature Package

Step 18A creates a standardized JSON representation of the measurements generated by the analyzer. This separates the DSP engine from the AI layer and provides a reproducible input for downstream interpretation.

Output	Purpose
reports\music_feature_summary.json	AI-ready structured audio feature package
src\music_feature_summary.py	Collects and packages Steps 4–17 measurements

11. Step 18B — AI Music Interpretation

Step 18B uses the structured feature package as input to Gemini AI. The AI is instructed to use only supplied measurements, avoid inventing unsupported track information, distinguish measured facts from interpretation, and return structured JSON.

Output	Purpose
reports\music_ai_interpretation.json	Structured AI interpretation with source measurements
reports\music_ai_interpretation.txt	Human-readable AI music interpretation
src\ai_music_understanding.py	Gemini-powered interpretation engine

12. Verified AI Interpretation

The verified Gemini interpretation described the track as a monophonic 240.02-second, 44.1 kHz file with -17.05 dBFS RMS and -0.17 dBFS peak. It identified a strongly bass-dominated spectrum (89.42% bass), moderate tempo (109.96 BPM), estimated G Minor key (0.7472 correlation), zero digital clipping and high technical integrity under the project's 96/100 health heuristic. The AI avoided unsupported claims about artist, genre, instruments, vocals and lyrics.

13. Professional Visualization Outputs

Output	Purpose
airdopes_waveform.png	Recorded Airdopes waveform
airdopes_audio_level.png	Signal-level visualization
airdopes_frequency_spectrum.png	FFT spectrum
airdopes_spectrogram.png	Time-frequency visualization
airdopes_spectral_features.png	Spectral feature visualization
airdopes_bass_mid_treble.png	Frequency-band energy
airdopes_tempo_beats.png	Tempo and beat timeline
music_analysis_dashboard_v2.png	Professional music dashboard
music_key_chroma.png	Key/chroma visualization
music_loudness_dynamic_range.png	Loudness and clipping visualization
audio_quality_health_score.png	Step 17 health-score visualization

14. Current Project Structure

```
Project-02-AI-Music-Audio-Analyzer/  
■■■ audio/ → airdopes_test.wav  
■■■ music/ → test_music.mp3  
■■■ reports/ → DSP graphs, dashboards, JSON and AI interpretation outputs  
■■■ src/ → Steps 1–18B Python scripts  
■■■ tests/  
■■■ venv/  
■■■ requirements.txt  
■■■ .gitignore  
■■■ README.md
```

15. Overall Validation

Validation	Status
Airdopes Bluetooth microphone accessible	PASS
Airdopes recording	PASS
WAV signal analysis	PASS
Waveform / FFT / spectrogram	PASS
Spectral features	PASS
Bass/Mid/Treble	PASS
Tempo/beat analysis	PASS
Integrated analyzer	PASS

Validation	Status
Real music input and analysis	PASS
Professional dashboard	PASS
Key/chroma estimation	PASS
Loudness/dynamic-range analysis	PASS
Clipping detection	PASS
Audio Quality & Health Score	PASS
AI-ready feature package	PASS
Gemini AI interpretation	PASS
AI Music Report / Dashboard Integration	NEXT

16. Security & Reproducibility

The Gemini API key is supplied through the GEMINI_API_KEY environment variable and is not embedded in Python source. API secrets, .env files and key files should remain excluded from GitHub. The AI layer consumes a generated JSON feature package so measured inputs can be inspected independently of the AI response.

17. Engineering Limitations

DSP measurements depend on selected algorithms, sample rate, mono conversion and frequency-band definitions. Key detection is an estimate based on chroma/profile correlation. The Audio Quality & Health Score is project-specific. Gemini output is an AI-generated explanation of measured features and is not a replacement for professional audio engineering or mastering analysis.

18. Next Development Step

Step 18C — AI Music Report & Dashboard Integration: combine DSP results, health score, visualizations and Gemini interpretation into one professional end-to-end music analysis dashboard/report.

19. Version History

v0.1 — Project-02 documentation baseline created from the beginning.

v0.2 — Steps 4–6: WAV analysis, waveform visualization and audio-level analysis.

v0.3 — Steps 7–8: FFT spectrum and STFT spectrogram.

v0.4 — Step 9: spectral feature analysis.

v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat analysis.

v0.6 — Steps 12–13: integrated feature summary and real music track analysis.

v0.7 — Step 14: professional music visualization dashboard and refined v2 layout.

v0.8 — Steps 15–16: key/chroma analysis plus loudness, dynamic range and clipping analysis.

v0.9 — Step 17: transparent Audio Quality & Health Score; final result 96/100 — Excellent.

v1.0 — Step 18A: AI-ready feature package; Step 18B: Gemini AI Music Interpretation with structured JSON and human-readable report.

Only Project-02 development history is included.